



Figure 1: The Proxmox installer: Three easy steps to deploy the virtualisation environment

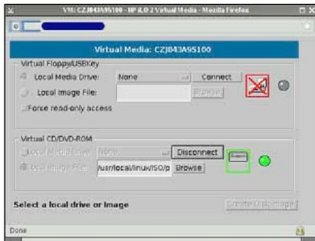


Figure 2: Attach the ISO image as virtual media to boot from

Installation

The Proxmox installer is rather simple, and provides no special features. For example, at the moment, it can partition disk storage as one LVM volume only (while leaving 500 MB as a normal ext3 partition for `/boot`). So I decided to install pure Debian first, and create a partition layout I considered correct for my purposes. After that, I installed the Proxmox packages. Luckily, this option works. Generally speaking, the default parameters could fit almost everyone's purposes, so don't be afraid to install Proxmox with its own installer.

I installed Proxmox on an HP ProLiant DL360 G5 with 18 GB RAM, two Quad-Xeon E5440s, and a Smart Array P400i with 360 GB of RAID 5 storage. Instead of RAID 5, you can rebuild it with a faster operational mode, like 'mirror'.

I attached a Debian Lenny ISO image to a virtual CD ROM in iLO2 (integrated Lights Out—the HP management console integrated into the hardware system) and bootstrapped the system (Figure 2).

During installation, I split my disk storage as follows: `'/'` partition occupies 19 GB, followed by 8 GB of swap space,

and the remaining space reserved for the LVM pool. You might wonder why I did it this way. First, in case of disk failure on the host (rather unlikely, but still...), I'm sure I'll get my configuration data back, with the help of a LiveCD, or forensic tools. Despite the huge number of LiveCDs, I frankly doubt they can help me mount a destroyed LVM volume on top of RAID 0/1/5. So if the LVM subsystem decides to die, the basic root and swap partitions will survive—yes, I'm this paranoid. And my final argument: when you talk about LVM, it means you're constructing something complex—say, logical volumes with mirror capabilities, or volumes residing on different physical disks. I try to follow Occam's razor rule, and leave LVM to jobs where it excels, while other duties should be carried out by other Linux components.

Next step: I installed the basic Debian packages (the base system, LVM and OpenSSH), assigned an IP address to the box, and added several lines to `/etc/apt/sources.list`:

```
# PVE packages provided by proxmox.com
deb http://download.proxmox.com/debian lenny pve
# Mirror site
deb http://ftp.dk.debian.org/debian/ lenny main non-free
# security updates
deb http://security.debian.org/ lenny/updates main
```

Next comes the usual update procedure:

```
# aptitude update && aptitude safe-upgrade
```

```
W: GPG error: http://download.proxmox.com lenny Release: The
following signatures couldn't be verified because the public key
is not available: NO_PUBKEY C23AC7F49887F95A
```

Unfortunately, I forgot to import the public key from the Proxmox repository, so let us fix it:

```
# gpg --keyserver pgpkeys.mit.edu --recv-key C23AC7F49887F95A
gpg: requesting key 9887F95A from hkp server pgpkeys.mit.edu
gpg: /root/.gnupg/trustdb.gpg: trustdb created
gpg: key 9887F95A: public key "Proxmox Release Key <proxmox-
release@proxmox.com>" imported
gpg: Total number processed: 1
gpg: imported: 1
# gpg -a --export C23AC7F49887F95A | apt-key add -
OK
```

Run `'aptitude update'` again to synchronise the local repository, then search for Proxmox packages available for install:

```
# aptitude search proxmox
p proxmox-ve-2.6.18
The Proxmox Virtual Environment
```